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Disturbance Observer-based Attitude Control for a Quadrotor

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***Abstract*—This paper gives an disturbance observer-based attitude control method of a quadrotor Unmanned Aerial Vehicle (UAV) in the presence of large disturbance. Based on linear PID controller a disturbance observer is designed to estimate and compensate for the impact of internal/external disturbance simultaneously. After the disturbance observer is added, the attitude subsystem can overcome the strong coupling and external disturbance. Simulation results show that the controller is simple and effective which is able to ensure the stability of attitude and has strong robustness to disturbance.**

Keywords—QBall2 Quadrotor; Disturbance Observer; Attitude Control; PID Control

I. INTRODUCTION

In recent years, because the quadrotor unmanned UAV has the characteristics of vertical take-off and landing, freedom to hover, simple structure, its application in the field of military, civil and scientific research is more and more widely. This trend is also partly due to the integration and application on UAV platforms of high performance processors, sensors and micro electro mechanical systems devices with lower and lower power consumption, which promote the development of quadrotor UAV [1]. To ensure that the quadrotor has good flying quality in a variety of flight conditions, good flight control is necessary in addition to good hardware equipment. So that reliable and stable flight control algorithm is very important. And for a quadrotor the attitude control is the basis which is one of the key technologies [2, 3]. As quadrotor is a strong coupling, multivariable, underactuated nonlinear system that has six degree of freedom ouputs and four inputs, its flight control also has received wide attention of researchers both at home and abroad. And at present all kinds of control algorithms are widely used in quadrotor attitude control, such as PID[4], LQR[5] linear control algorithm, feedback linearization[6], backstepping[7], sliding mode variable structure control[8] nonlinear control algorithm, and adaptive[9], H-infinity control[10], fuzz control[11] and so on. Disturbance and uncertainty rejection is a key objective in flight control system design. Some of these control approach mentioned above generally achieve the goal of disturbance rejection via feedback regulation based on the tracking error between the measured outputs and their set points. Thus, designed controllers cannot react directly and fast enough in the presence of strong disturbances relative to the feedforward control. However, quite often, the external disturbance cannot be directly measured or is too expensive to measure [12]. So an active disturbance-rejection controller for the flight control was given in [13] which designed an extended state observer to estimate and compensate for the impact of disturbance. However, the controller has too many parameters to adjust which is a little complicated. As one of the most effective and popular disturbance estimation techniques disturbances observer has attracted a lot of interest and disturbance observer-based control (DOBC) has received a great deal of attention which is applied in this paper. For the attitude stabilization problem of a quadrotor, a classical linear PID is used as the basic controller. And on this basis, the output of the PID controller is combined with a disturbance observer to estimate equivalent input disturbance in the system. With not join disturbance observer, a disturbance observer-based control is verified, tested the perturbation resistance of the system and robustness.

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II. DYNAMIC MODEL OF QBALL2 QUADROTOR

A quadrotor is a highly nonlinear, strong coupling of multiple inputs multiple outputs of complex nonlinear systems.

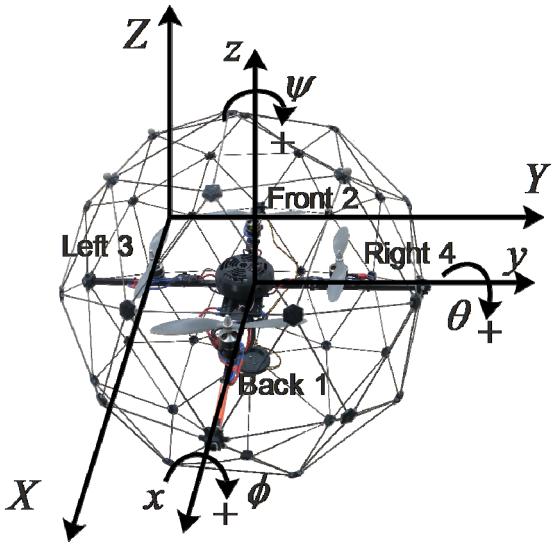

Fig. 1. The QBall2 quadrotor UAV .

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This paper is studied based on QBall2 quadrotor as shown in Fig. 1. The body axes coordinate system and the earth coordinate system of QBall2 are denoted (x, y, z) and (X, Y, Z) respectively. And the attitude of quadrotor is described using Euler angle. ϕ is roll angle and θ is pitch angle and ψ is yaw angle, which are defined as the angles of rotation about the *x*, *y* and *z* axis, respectively.

The mathematical model of QBall2 quadrotor is shown as (1) according to Newton-Euler formalism [14].

$$\begin{cases} \ddot{x} = \left[\left(\cos\phi \sin\theta \cos\psi + \sin\phi \sin\psi \right) U_1 \right] / m \\ \ddot{y} = \left[-\left(\cos\phi \sin\theta \sin\psi - \sin\phi \cos\psi \right) U_1 \right] / m \\ \ddot{z} = \left[\left(\cos\phi \cos\theta \right) U_1 \right] / m - g \\ \ddot{\phi} = \left[U_2 + \dot{\theta}\dot{\psi} \left(I_y - I_z \right) + d_{\Gamma\phi} \right] / I_x \\ \ddot{\theta} = \left[U_3 + \dot{\phi}\dot{\psi} \left(I_z - I_x \right) + d_{\Gamma\theta} \right] / I_y \\ \ddot{\psi} = \left[U_4 + \dot{\phi}\dot{\theta} \left(I_x - I_y \right) + d_{\Gamma\psi} \right] / I_z \end{cases} \tag{1}$$

where *m* is the total mass of quadrotor; *g* is the acceleration of gravity; I_x, I_y, I_z are the moment of inertia about the *x*-axis, *y*-axis, *z*-axis respectively. U_1, U_2, U_3, U_4 are the virtual control inputs of quadrotor and U_1 denotes the total lift of a quadrotors and U_2, U_3, U_4 denote roll, pitch, yaw moment respectively.

The mathematical model of the quadrotor can be divided into position subsystem and attitude subsystem. This paper mainly studies the control problem of attitude angle under the condition of disturbance exists. And for the attitude subsystem in (1) $d_\Gamma = \left[d_{\Gamma\phi}, d_{\Gamma\theta}, d_{\Gamma\psi} \right]$ denotes attitude disturbance torque, on behalf of the airflow disturbance torque of quadrotor and they have no clear model information which belongs to the structural interference, but are slowly time-varying bounded signal.

The thrust T_i and un-torque Q_i generated by ith rotor have the following approximate relationship with ith motor's control input signal u_i as shown in (2)

$$T_i = K_t \frac{\omega}{s+\omega} u_i, Q_i = K_y \frac{\omega}{s+\omega} u_i, \quad (i = 1, 2, 3, 4) \tag{2}$$

where K_t and K_y are positive gains which denote thrust coefficient and un-torque coefficient respectively. ω is the motor bandwidth and u_i is the PWM control input of ith motor which is limited between 0-1.

And the virtual control inputs of a quadrotor have the following relationship with the thrust and un-torque generated by rotors as shown in equation (3) where *l* is the distance between the center of a rotor and the quadrotor.

$$\begin{cases} U_1 = \sum_i^4 T_i \\ U_2 = \left(T_1 - T_2 \right) l \\ U_3 = \left(T_1 - T_2 \right) l \\ U_4 = Q_1 - Q_2 + Q_3 - Q_4 \end{cases} \tag{3}$$

III. Disturbacnce Observer-Based Attitude Control Design

A. Disturbance Observer Design

For roll angle subsystem shown in (4)

$$I_x \ddot{\phi} = \dot{\theta}\dot{\psi} \left(I_y - I_z \right) + U_2 + d_{\Gamma\phi} + d_{\Delta\phi} \tag{4}$$

where part of the uncertainty of roll angle system model $d_{\Delta\phi}$ is considered and is combined with disturbance torque $d_{\Gamma\phi}$ as the total disturbance d_ϕ of roll angle as shown in (5).

$$d_\phi = d_{\Gamma\phi} + d_{\Delta\phi} \tag{5}$$

So equation (4) can be rewritten in

$$\ddot{\phi} = \left[\dot{\theta}\dot{\psi} \left(I_y - I_z \right) + U_2 + d_\phi \right] / I_x \tag{6}$$

Then rewrite (6) in the form of affine nonlinear system

$$x = f(x) + g_1(x)u + g_2(x)d \tag{7}$$

So equation (6) can be rewritten in

$$\ddot{\phi} = \frac{\dot{\theta}\dot{\psi} \left(I_y - I_z \right)}{I_x} + \frac{1}{I_x} U_2 + \frac{1}{I_x} d_\phi \tag{8}$$

To estimate the disturbance, a nonlinear disturbance observer is depicted by the following equation.

$$\begin{cases} \dot{z} = -l(x) g_2(x) z - \\ \qquad l(x) \left[g_2(x) p(x) + f(x) + g_1(x) u \right] \\ \hat{d} = z + p(x) \end{cases} \tag{9}$$

And the nonlinear distance observer gain $l(x)$ is determined by:

$$l(x) = \frac{\partial p(x)}{\partial x} \tag{10}$$

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So the disturbance observer of roll angle is show as equation (11).

$$\begin{cases} \dot{z} = -l(x) * z / I_x - \\ \quad l(x) * \left[p(x) + \dot{\theta}\dot{\psi}\left(I_y - I_z\right) + U_2 \right] / I_x \\ \hat{d} = z + p(x) \end{cases} \tag{11}$$

$$p(x) = l(x) * \dot{\phi}$$

The disturbance estimation error is defined as.

$$e_d = d - \hat{d} \tag{12}$$

And assuming that disturbance is slowly time-varying signal.

$$\dot{d} = 0 \tag{13}$$

So the derivative of (12) is

$$\dot{e}_d = \dot{d} - \dot{\hat{d}} \tag{14}$$

Combing (13) and (14):

$$\begin{aligned} \dot{e}_d &= 0 - \left(\dot{z} + \dot{p}(x)\right) \\ &= -\dot{z} - l(x) * \ddot{\phi} \\ &= l(x) * z / I_x + l(x) * [p(x) + \dot{\theta}\dot{\psi}\left(I_y - I_z\right) \\ &\quad + U_2] / I_x - l(x) * \ddot{\phi} \end{aligned} \tag{15}$$

Replacing $\ddot{\phi}$ by (8)

$$\begin{aligned} \dot{e}_d &= l(x)z / I_x + l(x)\left[p(x) + \dot{\theta}\dot{\psi}\left(I_y - I_z\right) + U_2 \right] / I_x \\ &\quad - l(x)\left(\frac{\dot{\theta}\dot{\psi}\left(I_y - I_z\right)}{I_x} + \frac{1}{I_x}U_2 + \frac{1}{I_x}d \right) \\ &= l(x)\left(z + p(x)\right) / I_x - l(x)d / I_x + l(x)[\dot{\theta}\dot{\psi}(I_y - \\ &\quad I_z) + U_2] / I_x - l(x)\left(\frac{\dot{\theta}\dot{\psi}\left(I_y - I_z\right)}{I_x} + \frac{1}{I_x}U_2 \right) \\ &= l(x)\hat{d} / I_x - l(x)d / I_x \\ &= -l(x)e_d / I_x \end{aligned} \tag{16}$$

Which implies that the disturbance estimation error will converge to zero as time goes to infinity if the observer gain $l\left(x\right)$ is chosen such that system (16) is asymptotically stable.

B. Controller Input Design

For the basic attitude control, a PID controller is used for roll angle and pitch angle. Due to the weak control torque of yaw and usually not very strict for yaw, a PD controller is used for yaw angle control. The control signals combined with estimated disturbance are given to four motor as shown in equation (17).

$$\begin{cases} u_1 = u_throttle + u_pitch - u_yaw \\ u_2 = u_throttle - u_pitch - u_yaw \\ u_3 = u_throttle + u_roll + u_yaw \\ u_4 = u_throttle - u_roll + u_yaw \end{cases} \tag{17}$$

where *u_throttle* is throttle control signal; *u_roll* is roll angle control signal; *u_pitch* is pitch angle control signal; *u_yaw* is yaw angle control signal.

IV. Simulation and Results

The observer disturbance-based attitude control structure is as shown in Fig. 2.

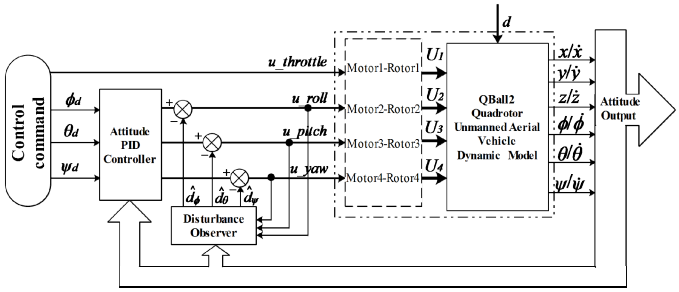

Fig. 2. The control system structure block diagram of QBall2.

where ϕ_d, θ_d, ψ_d are desired attitude angle; d is the disturbance for the QBall2 quadrotor and $\hat{d}_\phi, \hat{d}_\theta, \hat{d}_\psi$ are the disturbance estimated by disturbance observer. And the model parameters of QBall2 quadrotor are shown in Table 1.

TABLE I. MODEL PARAMETERS OF QBALL2

Symbol	Value
m	1.80kg
l	0.20m
w	15rad/s
I_x,	0.03kg*m^2
I_y	0.03kg*m^2
I_z	0.04kg*m^2
K_t	8.80N
K_y	0.40N
g	9.81m/(s^2)

The simulation time is set to 100s, assuming that quadrotor take off first then desired attitude is given at 15s. And in the 20s disturbance is given which is slowly time-varying

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aerodynamic interference torque and some system uncertainty is included in as shown in equation (18). The observer gains $l(x)$ are chosen as $l(x)=[0.1,0.1,1]$ and disturbance are chosen as

$$\begin{aligned} d &= [d_\phi, d_\theta, d_\psi] \\ &= [0.1\sin(0.1\pi t)+0.1, 0.1\cos(0.1\pi t)+0.1, \\ &\quad 0.01\sin(0.1\pi t)+0.02] \end{aligned} \tag{18}$$

The simulation results are shown in the following.

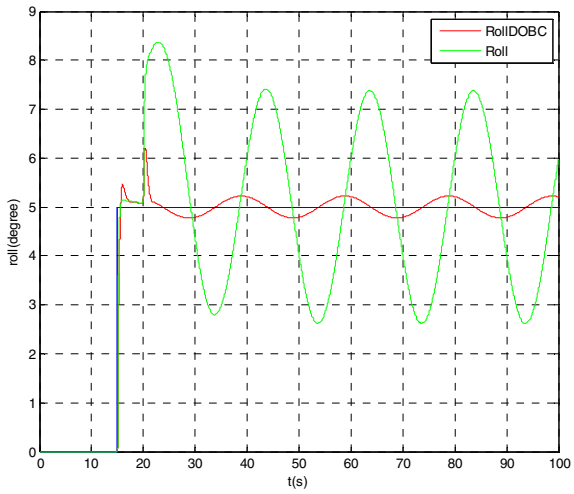

Fig. 3. Roll angle output.

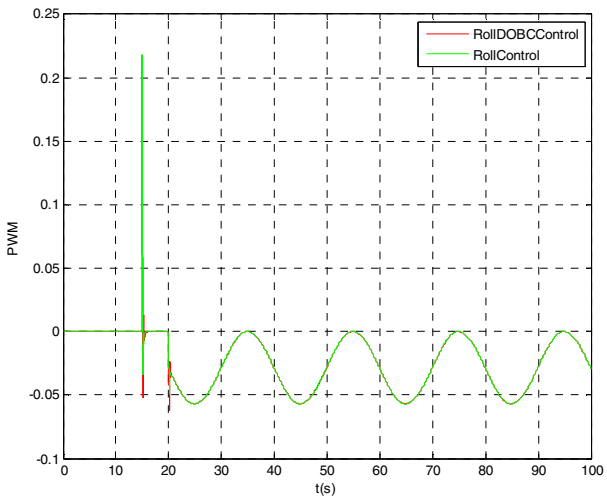

Fig. 4. Roll angle control input signal.

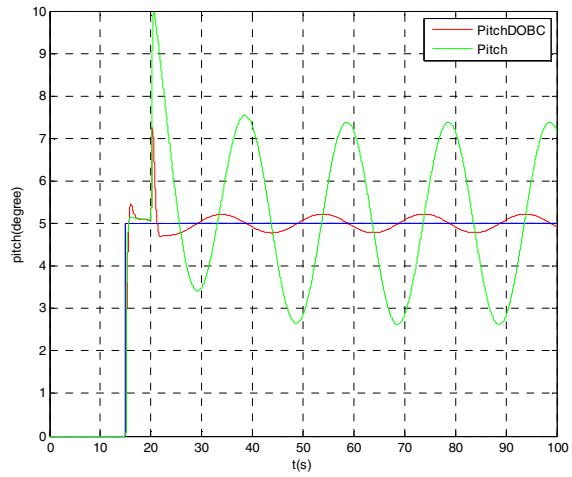

Fig. 5. Pitch angle output.

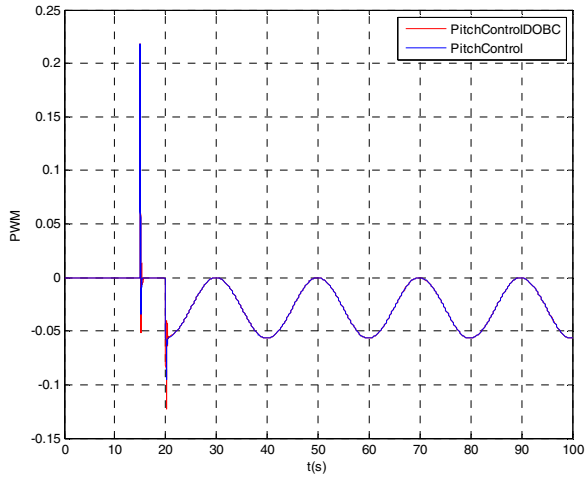

Fig. 6. Pitch angle control input signal.

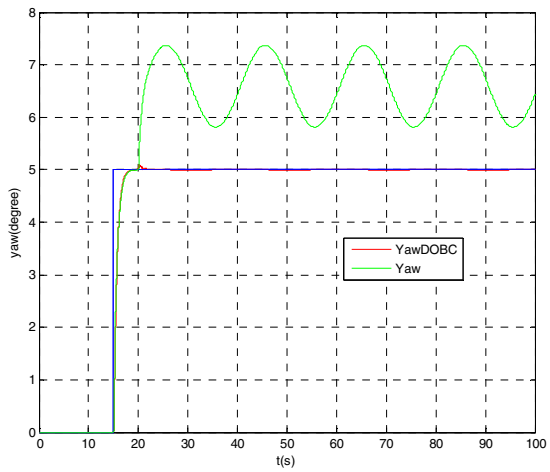

Fig. 7. Yaw angle output.

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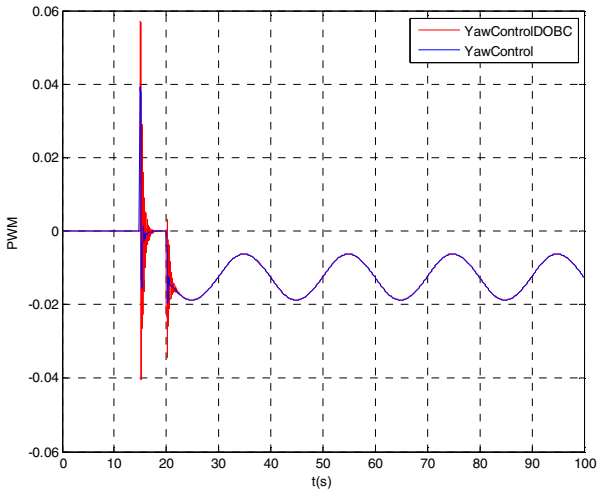

Fig. 8. Yaw angle control input signal.

Fig. 3 to Fig. 8 show that disturbance observer-based attitude control has better robustness and need nearly the same control input signal than traditional PID controller.

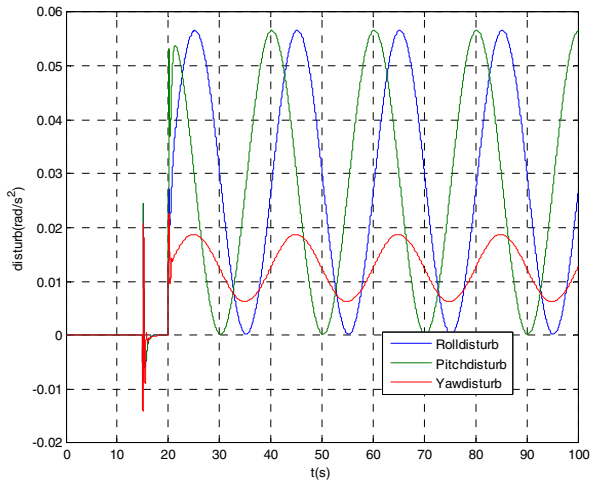

Fig. 9. Disturbacne observered by disturbance obeserver.

Fig. 9 shows that the disturbance observer designed has a good performance to estimate the system disturbance.

White noise is added to the attitude angles of quadrotor to represent internal disturbance which caused by sensors noise.

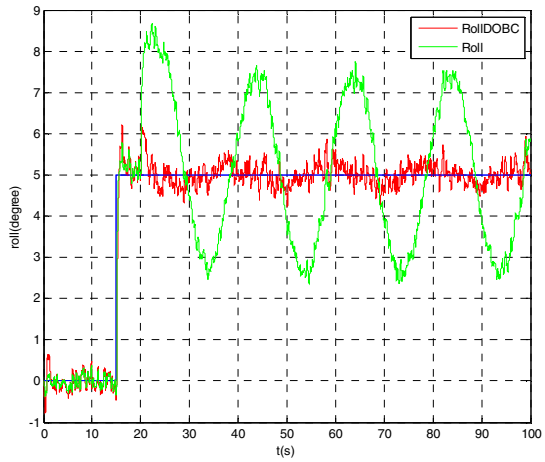

Fig. 10. Roll angle output with sensors noise.

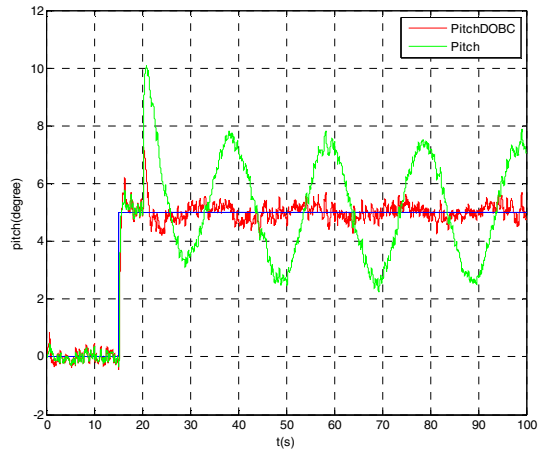

Fig. 11. Pitch angle output with sensors noise.

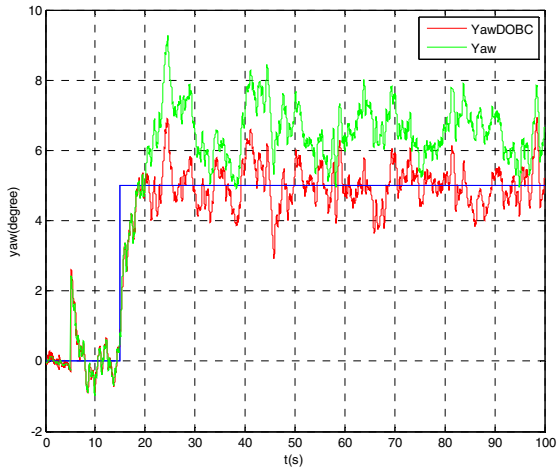

Fig. 12. Yaw anglewith sensors noise.

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Fig.10. to Fig. 12. show that the disturbance observer-based attitude control can still has a good robustness in the case of the high frequency oscillation disturbance cause by the vibration of the quadrotor flight..

V. CONCLUSIONS

For underactuated quadrotor a nonlinear disturbance observer is designed in this paper which make attitude subsystem fast track command signal, and overcome the aerodynamic disturbance torque and model parameter uncertainty. Theoretical analysis and simulation results show that the designed controller can improve the anti-interference ability of the flight control system, and disturbance observer-based control method can enhance the robustness and does not need to change the existing controller structure only require the system to control input and system output to estimate the equivalent interference which is simple and convenient, has extensive adaptability.

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